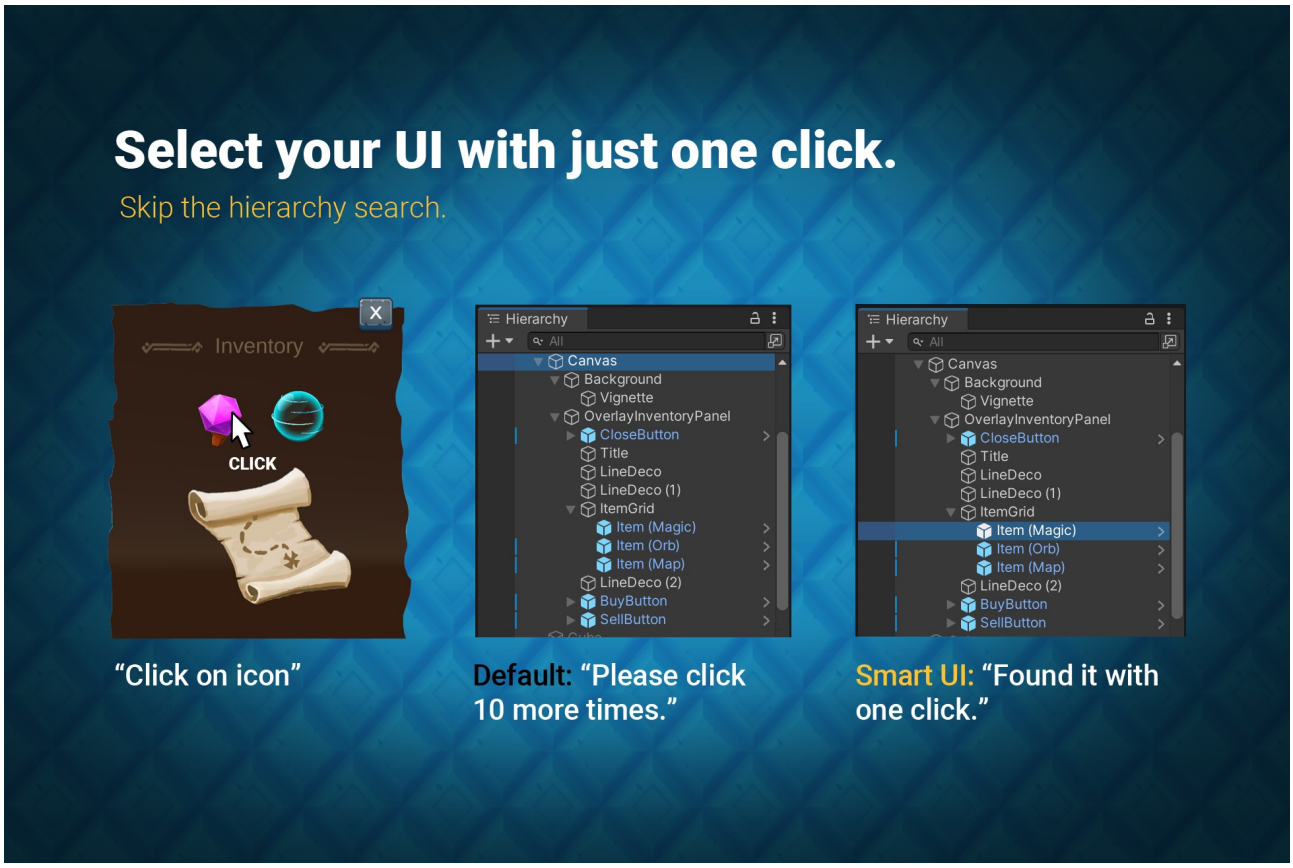
Smart UI Selection Manual

for Unity 6, 2021, 2022 and 2023



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Requirements & Setup

Requirements

Unity 2019.4 or higher is required. Yes indeed it does support Unity versions from 2019 up to Unity 6. Though future versions may drastically reduce support for older versions.

Overview

IMPORTANT: If you want to keep your settings when upgrading to a new version. Then please uncheck the SmartUISelection_Settings.asset during import.

This is an EDITOR ONLY plugin.

This plugin enhances the ui selection functionality in the SceneView of the Unity Editor.

It improves the selection by ignoring empty ui elements and allows for click-through selection of 3D objects behind the canvas.

It also fixes the (sometimes) random order of ui selections done by the Editor.

It also tries to address the issue that the screen space overlay canvases are quite often in the way of selecting anything in the 3D scene.

It does so by disabling those canvases if you get very close to them.

This feature is disabled by default to avoid confusing for new users ("help my canvas is gone").

You can easily turn it on in the settings (Tools -> Smart UI Selection -> Turn Auto Hide On).

Support

If you find any errors then please write to office@kamgam.com.

Please be patient, replies may take a few days. Please include

your Asset Store Order Nr in your request.

Thank you.

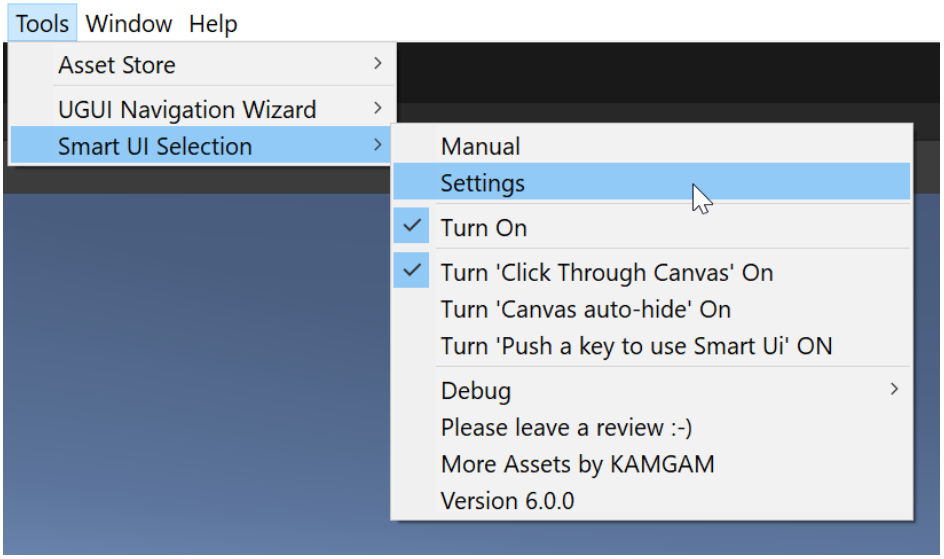
Known Issues / Wishlist

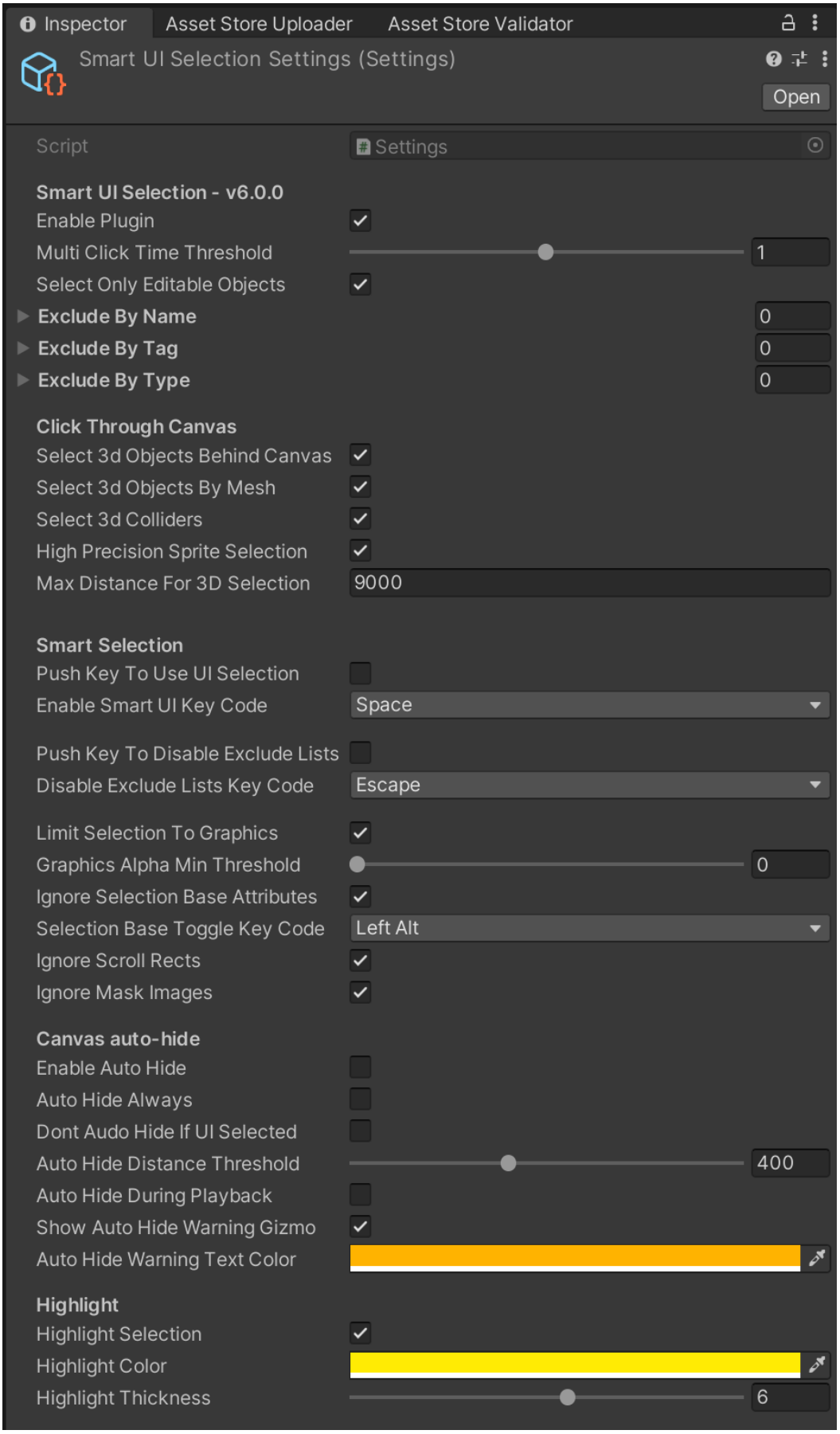
* The Click-through detection does not recognize if a texture on a Graphic component has transparent pixels. At the moment only alpha values of colors, renderers or canvas groups are recognized.

Settings

You can find or recreate the settings file (named "SmartUISelection Settings") through the main menu (Tools > SmartUISelection > Settings).

You can configure some of the most commonly used setting directly through the Tools > .. menu:




Inspector
Asset Store Uploader
Asset Store Validator
Smart UI Selection Settings (Settings)
Open
Script
Settings
Smart UI Selection - v6.0.0
Enable Plugin
Multi Click Time Threshold
1
Select Only Editable Objects
Exclude By Name
0
Exclude By Tag
0
Exclude By Type
0
Click Through Canvas
Select 3d Objects Behind Canvas
Select 3d Objects By Mesh
Select 3d Colliders
High Precision Sprite Selection
Max Distance For 3D Selection
9000
Smart Selection
Push Key To Use UI Selection
Enable Smart UI Key Code
Space
Push Key To Disable Exclude Lists
Disable Exclude Lists Key Code
Escape
Limit Selection To Graphics
Graphics Alpha Min Threshold
0
Ignore Selection Base Attributes
Selection Base Toggle Key Code
Left Alt
Ignore Scroll Rects
Ignore Mask Images
Canvas auto-hide
Enable Auto Hide
Auto Hide Always
Dont Audo Hide If UI Selected
Auto Hide Distance Threshold
400
Auto Hide During Playback
Show Auto Hide Warning Gizmo
Auto Hide Warning Text Color
Highlight
Highlight Selection
Highlight Color
Highlight Thickness
6

* Enable Plugin

Enables or disables the whole plugin. Turn this off if you suspect the plugin to cause any weird behaviour (none known at the moment).

* Multi Click Time Threshold

If you click twice within this timeframe then the selection will cycle through all found elements. Time is in seconds.

* Select Only Editable Objects

If checked then objects whose hideFlags are set to HideFlags.NotEditable will be ignored.

CLICK THROUGH CANVAS

* Select 3D Objects Behind Canvas

If no ui element has been selected then try to click through the canvas and select 3D objects behind it. Uses the 3d objects colliders or a bounding box if there is no collider.

* Select 3D Objects By Mesh

This may slow down the click handling (if clicked through a canvas in big scenes). Enabling this will make the click-through feature more accurate for meshes without colliders. It will do a raycast on all triangles of each mesh (even those without a collider).

* Select 3D Colliders

Select 3D objects based on their colliders too. Useful for invisible objects which solely consist of colliders (like a trigger).

Usually 3D objects are only selected based on their mesh.

* High Precision Sprite Selection

Turn off if selection is slow in scenes with a lot of SpriteRenderers.

If turned on then clicks are checked against the actual sprite mesh.

* Max Distance For 3D Selection

A raycast is used to detect 3D objects. This sets the maximum distance for the raycast in world units.

SMART UI SELECTION

* Push Key To Use Ui Selection

If checked then Smart UI Selection is only enabled if you press the SPACE key.

* Enable Smart UI Key Code

Push and HOLD this key to disable Smart UI Selection (works only if "Push Key To Use Ui Selection" is turned on).

* Push Key To Disable Exclude Lists

If checked and if the key is pressed then Smart UI Selection will ignore your exclude lists (act as if they are empty).

* Disable Exclude Lists Key Code

Push and HOLD this key to disable the exclude lists (works only if "Push Key To Disable Exclude Lists" is turned on).

* Limit Selection To Graphics

Limit ui selection to elements with graphics (objects which have a 'Graphic' component).

* Alpha Threshold

Select elements only if they have an alpha value above the threshold. 'Limit Selection To Graphics' needs to be turned on for this to have any effect.

* Ignore SelectionBase Attributes

Check to completely ignore the [SelectionBase] Attributes.

* Selection Base Toggle Key Code

If that key is pressed the the behaviuor of „Ignore SelectionBase Attributes" is inverted. That's useful if you want to select the base prefab instead of the specific UI element.

* Exclude By Name

Add names of objects which should not be selectable. The names have to match exactly.

* Exclude By Tag

Add tags of objects which should not be selectable. The names have to match exactly.

* Ignore ScrollRects

Whether or not ScrollRects should be ignored.

* Ignore Mask Images

Whether or not the images used by Mask Components should be ignored.

* Highlight:

If highlights are enabled then the tool will show a big selection border around the selected elements in the scene view. Very handy if you routinely select multiple UI elements.

CANVAS AUTO-HIDE

* Enable Auto Hide?

Should ScreenSpaceOverlay canvases be hidden in the scene view if the editor camera gets very close?

Useful to prohibit unwanted canvas selections while you edit the 3d scene.

They will only be hidden if your mouse cursor is in the scene view.

Notice that this will only affect ScreenSpaceOverlay canvases.

* Auto Hide Always

If enabled then auto hide will always affect the canvases independently of where the mouse cursor is. Enabling this works only in Unity 2019.2+. I found this turned on to be the most helpful setting BUT use it with caution. New Smart UI Selection users may be confused by it because the only way to show the hidden canvases is to zoom out (thus it is disabled by default).

* Auto Hide Distance Threshold

ScreenSpaceOverlay canvases will be hidden if the editor camera distance to the XY plane is less than X.

* Auto Hide During Playback

Should ScreenSpaceOverlay canvases be hidden in play mode too?

* Show Auto Hide Warning Gizmo

Show a warning text next to canvases to indicate that auto-hide is turned on? Useful in team setups when others might not expect a canvas to automatically vanish in the SceneView.